

Introduction to Java(Ca-1 Practice question)

◆ History of Java

- Java was developed in **1991** by **James Gosling** and his team at **Sun Microsystems**
 - Earlier name: **Oak**
 - Main goal: *Write Once, Run Anywhere*
 - Later acquired by **Oracle Corporation**
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◆ Features of Java

1. **Simple** – Easy syntax, similar to C/C++
 2. **Object-Oriented** – Everything is based on objects and classes
 3. **Platform Independent** – Java code runs on any OS
 4. **Secure** – No direct memory access
 5. **Robust** – Strong memory management
 6. **Multithreaded** – Multiple tasks at same time
 7. **High Performance** – Uses JIT compiler
 8. **Portable** – Bytecode runs anywhere
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◆ Java Program Structure

A Java program has:

1. Package statement (optional)
2. Import statement (optional)
3. Class definition
4. main() method
5. Statements inside main

Example:

```
class Hello {  
    public static void main(String[] args) {  
        System.out.println("Hello Java");  
    }  
}
```

```
}  
}
```

◆ Writing Simple Java Class & main() Method

- **class** → Blueprint
- **main()** → Starting point of program
- JVM always starts execution from main()

Syntax:

```
public static void main(String[] args)
```

◆ Command-Line Arguments

- Used to pass data while running program
- Stored in args[]

Example:

```
class Test {  
    public static void main(String[] args) {  
        System.out.println(args[0]);  
    }  
}
```

Run:

```
java Test Hello
```

Output:

```
Hello
```

◆ JDK, JRE, JVM

JVM (Java Virtual Machine)

- Runs Java bytecode
- Platform dependent

JRE (Java Runtime Environment)

- JVM + Libraries
- Used to **run** Java programs

JDK (Java Development Kit)

- JRE + Compiler + Tools
- Used to **develop** Java programs

👉 **JDK > JRE > JVM**

Data in the Cart (Core Concepts)

◆ Primitive Data Types

Type	Size	Example
byte	1 byte	10
short	2 bytes	100
int	4 bytes	1000
long	8 bytes	10000L
float	4 bytes	10.5f
double	8 bytes	10.55
char	2 bytes	'A'
boolean	1 bit	true

◆ Type Conversion

1 Implicit (Widening)

```
int a = 10;  
double b = a;
```

2 Explicit (Narrowing)

```
double a = 10.5;  
int b = (int)a;
```

◆ Keywords

- Reserved words in Java
 - Cannot be used as variable names
Examples: int, class, static, public, if
-

◆ Identifiers

- Names of variables, classes, methods
- Must not start with number
- Cannot use keywords

Valid:

`int totalMarks;`

Invalid:

`int 1num;`

◆ Variables

Types:

1. **Local variable**
 2. **Instance variable**
 3. **Static variable**
-

◆ Access Modifiers

Modifier	Access
----------	--------

public	Everywhere
--------	------------

protected	Same package + subclass
-----------	-------------------------

default	Same package
---------	--------------

private	Same class
---------	------------

◆ **static Keyword**

- Belongs to class, not object
 - Used for memory saving
 - `main()` is static
-

◆ **Wrapper Classes**

- Convert primitive → object

Primitive Wrapper

`int` Integer

`char` Character

`double` Double

`boolean` Boolean

Example:

Integer a = 10;

⚡ **Operators**

◆ **Arithmetic Operators**

`+` `-` `*` `/` `%`

◆ **Relational Operators**

`<` `>` `<=` `>=` `==` `!=`

◆ **Logical Operators**

`&&` `||` `!`

◆ **Bitwise Operators**

`&` `|` `^` `~` `<<` `>>`

◆ Unary Operators

`++ --`

◆ Assignment Operators

`= += -= *=`

◆ Ternary Operator

`result = (a > b) ? a : b;`

◆ Operator Precedence

Highest → Lowest:

1. Unary
 2. Arithmetic
 3. Relational
 4. Logical
 5. Assignment
-

Conditional Statements

◆ if Statement

```
if(condition) {  
    // code  
}
```

◆ if-else

```
if(a > b) {  
    System.out.println("A is big");  
} else {
```

```
System.out.println("B is big");  
}
```

◆ **else-if Ladder**

```
if(marks > 90) grade = "A";  
else if(marks > 75) grade = "B";  
else grade = "C";
```

◆ **switch-case**

```
switch(day) {  
    case 1: System.out.println("Monday"); break;  
    case 2: System.out.println("Tuesday"); break;  
    default: System.out.println("Invalid");  
}
```

Top 30 MCQs (With Answers)

1. Java was developed by?
A) Microsoft
B) Apple
C) Sun Microsystems ☒
D) IBM
2. Java is a ____ language
A) Procedural
B) Object-Oriented ☒
C) Functional
D) Script
3. Entry point of Java program?
A) start()
B) main() ☒
C) init()
D) run()
4. Which makes Java platform independent?
A) Compiler
B) JVM ☒

- C) JDK
- D) JRE

5. Size of int?

- A) 2 bytes
- B) 4 bytes ☒
- C) 8 bytes
- D) 1 byte

6. Which is not primitive?

- A) int
- B) float
- C) String ☒
- D) char

7. Implicit conversion is also called?

- A) Narrowing
- B) Casting
- C) Widening ☒
- D) Boxing

8. Keyword used to define class?

- A) new
- B) define
- C) class ☒
- D) object

9. Which is a valid identifier?

- A) 2num
- B) num-1
- C) _num ☒
- D) int

10. Default access modifier?

- A) public
- B) private
- C) protected
- D) default ☒

11. static members belong to?

- A) Object
- B) Class ☒
- C) Method
- D) Package

12. Wrapper class of int?

- A) Int
- B) Integer ☒
- C) int
- D) Number

13. Which operator gives remainder?

- A) /
- B) % ☒
- C) *
- D) +

14. Logical AND operator?

- A) &
- B) && ☒
- C) ||
- D) !

15. Bitwise OR operator?

- A) ||
- B) &&
- C) | ☒
- D) ^

16. Unary operator example?

- A) +
- B) ++ ☒
- C) ==
- D) &&

17. Ternary operator symbol?

- A) ?: ☒
- B) ::
- C) ??
- D) ->

18. Which is conditional statement?

- A) for
- B) while
- C) if ☒
- D) break

19. switch works with?

- A) int

- B) char
- C) String
- D) All ☒

20. Which is not access modifier?

- A) public
- B) private
- C) static ☒
- D) protected

21. JVM converts bytecode into?

- A) Source code
- B) Machine code ☒
- C) Assembly
- D) Binary

22. JRE is used for?

- A) Development
- B) Execution ☒
- C) Compilation
- D) Editing

23. Which operator has highest precedence?

- A) Arithmetic
- B) Relational
- C) Unary ☒
- D) Assignment

24. Boolean values are?

- A) 0 and 1
- B) yes/no
- C) true/false ☒
- D) on/off

25. char size in Java?

- A) 1 byte
- B) 2 bytes ☒
- C) 4 bytes
- D) 8 bytes

26. Command line arguments stored in?

- A) args[] ☒
- B) input[]

- C) data[]
- D) main[]

27. Which keyword prevents inheritance?

- A) static
- B) final ☒
- C) private
- D) abstract

28. Which is NOT a loop?

- A) for
- B) while
- C) do-while
- D) if ☒

29. Java source file extension?

- A) .jav
- B) .class
- C) .java ☒
- D) .js

30. Which company owns Java now?

- A) Sun
- B) IBM
- C) Google
- D) Oracle ☒

LOOPS IN JAVA

Loops are used to **repeat a block of code** until a condition is satisfied.

◆ for Loop

Used when the number of iterations is **known**.

Syntax:

```
for(initialization; condition; increment/decrement) {  
    // code  
}
```

Example:

```
for(int i = 1; i <= 5; i++) {  
    System.out.println(i);  
}
```

◆ while Loop

Used when the number of iterations is **not fixed**.

Syntax:

```
while(condition) {  
    // code  
}
```

Example:

```
int i = 1;  
while(i <= 5) {  
    System.out.println(i);  
    i++;  
}
```

◆ do-while Loop

- Executes **at least once**, even if condition is false.

Syntax:

```
do {  
    // code  
} while(condition);
```

Example:

```
int i = 1;  
do {  
    System.out.println(i);  
    i++;  
} while(i <= 5);
```

◆ for-each Loop

Used for **arrays and collections**.

Syntax:

```
for(dataType variable : array) {  
    // code  
}
```

Example:

```
int[] arr = {10, 20, 30};  
for(int x : arr) {  
    System.out.println(x);  
}
```

 ARRAYS AND ENUMS◆ **Arrays (Fundamentals)**

- Used to store **multiple values of same data type**
- Fixed size

Example:

```
int[] marks = {80, 90, 70};
```

◆ **Multi-Dimensional Arrays**

Array inside an array (matrix).

Example:

```
int[][] matrix = {  
    {1, 2},  
    {3, 4}  
};
```

◆ **Array Access and Iteration**

Access using index:

```
marks[0];
```

Using loop:

```
for(int i = 0; i < marks.length; i++) {  
    System.out.println(marks[i]);  
}
```

◆ Varargs (Variable Arguments)

Allows passing **multiple arguments** to a method.

Example:

```
void sum(int... a) {  
    int total = 0;  
    for(int x : a)  
        total += x;  
    System.out.println(total);  
}
```

◆ Enumerations (enum)

Used to define **fixed set of constants**.

Example:

```
enum Day { MON, TUE, WED }
```

```
Day d = Day.MON;
```

OOP CONCEPTS

◆ Class and Objects

- **Class** → Blueprint
- **Object** → Real-world entity

Example:

```
class Student {  
    int id;  
    String name;  
}
```

◆ Constructors

- Special method
- Same name as class
- No return type
- Used to initialize objects

Example:

```
class Student {  
    Student() {  
        System.out.println("Constructor called");  
    }  
}
```

◆ Methods

Block of code that performs a task.

```
void display() {  
    System.out.println("Hello");  
}
```

◆ Method Overloading

Same method name but **different parameters**.

```
int add(int a, int b) { return a + b; }  
int add(int a, int b, int c) { return a + b + c; }
```

◆ Constructor Overloading

More than one constructor in a class.

```
Student() {}  
Student(int id) {}
```

◆ this Keyword

Refers to **current object**.

```
this.id = id;
```

◆ Initializer Blocks

Executed **before constructor**.

Instance initializer:

```
{  
    System.out.println("Initializer block");  
}
```

STRING CLASS

◆ String Class

- Immutable (cannot be changed)

Constructors:

```
String s1 = "Java";  
String s2 = new String("Java");
```

Common Methods:

- length()
 - charAt()
 - toUpperCase()
 - toLowerCase()
 - equals()
 - substring()
 - concat()
-

◆ StringBuilder Class

- Mutable (can be changed)
- Faster than String

Example:


```
StringBuilder sb = new StringBuilder("Java");  
sb.append(" World");
```

Common Methods:

- append()
 - insert()
 - delete()
 - reverse()
-

TOP 30 MCQs (WITH ANSWERS)

1. Which loop executes at least once?
A) for
B) while
C) do-while ☒
D) for-each
2. for-each loop is used with?
A) Files
B) Arrays ☒
C) Threads
D) Classes
3. Array size in Java is?
A) Dynamic
B) Fixed ☒
C) Variable
D) Unlimited
4. Index of first element in array?
A) 1
B) -1
C) 0 ☒
D) Any
5. Multi-dimensional array is also called?
A) Linear
B) Jagged
C) Matrix ☒
D) Static

6. Which keyword is used for variable arguments?

- A) args
- B) var
- C) ... ☒
- D) array

7. enum is used to store?

- A) Variables
- B) Objects
- C) Constants ☒
- D) Methods

8. Class is a?

- A) Object
- B) Blueprint ☒
- C) Method
- D) Variable

9. Object is created using?

- A) class
- B) this
- C) new ☒
- D) static

10. Constructor name must be?

- A) Any name
- B) Class name ☒
- C) main
- D) init

11. Constructor has return type?

- A) Yes
- B) No ☒
- C) int
- D) void

12. Method overloading depends on?

- A) Return type
- B) Access modifier
- C) Parameters ☒
- D) Class name

13. this keyword refers to?

- A) Parent object

- B) Current object ☒
- C) Class
- D) Method

14. Initializer block runs?

- A) After main
- B) After method
- C) Before constructor ☒
- D) After constructor

15. String objects are?

- A) Mutable
- B) Immutable ☒
- C) Dynamic
- D) Static

16. Which class is mutable?

- A) String
- B) StringBuilder ☒
- C) Character
- D) Object

17. length() method belongs to?

- A) Array
- B) String ☒
- C) Loop
- D) Enum

18. append() method belongs to?

- A) String
- B) StringBuilder ☒
- C) Array
- D) Enum

19. Which loop is best when iterations are known?

- A) while
- B) do-while
- C) for ☒
- D) for-each

20. Array index out of range causes?

- A) Compile error
- B) Logical error

C) Runtime error ☒

D) Syntax error

21. for-each loop can modify array elements?

A) Yes

B) No ☒

C) Sometimes

D) Depends

22. Which is not OOP concept?

A) Encapsulation

B) Inheritance

C) Compilation ☒

D) Polymorphism

23. Constructor is called when?

A) Method invoked

B) Object created ☒

C) Class loaded

D) Program ends

24. Which is correct String creation?

A) String s = 'Java';

B) String s = "Java"; ☒

C) String s = Java;

D) String s = new char[];

25. reverse() method is in?

A) String

B) StringBuilder ☒

C) Array

D) Object

26. enum constants are?

A) Variables

B) Objects

C) Public static final ☒

D) Private

27. Varargs can accept?

A) Only one value

B) Two values

C) Zero or more values ☒

D) Only arrays

28. Which loop is entry-controlled?

- A) do-while
- B) while
- C) for
- D) for & while ☒

29. Constructor overloading is an example of?

- A) Inheritance
- B) Polymorphism ☒
- C) Encapsulation
- D) Abstraction

30. String comparison should be done using?

- A) ==
- B) equals() ☒
- C) compare
- D) match